Practical No : 10

Consider a disk queue with requests for I/O to blocks on cylinders 98, 183 41, 122, 14, 124, 65, 67. The SSTF scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199.Write a Program to calculate the total head movement (in number ofcylinders) incurred while servicing these requests. Give me c code

INPUT :

```
  GNU nano 8.7                                                    sstf.c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int requests[] = {98, 183, 41, 122, 14, 124, 65, 67};
    int n = 8;
    int head = 53;
    int total_movement = 0;
    int visited[8] = {0};

    printf("Order of servicing:\n");

    for(int i = 0; i < n; i++) {
        int min = 1000;
        int index = -1;

        for(int j = 0; j < n; j++) {
            if(!visited[j]) {
                int distance = abs(head - requests[j]);
                if(distance < min) {
                    min = distance;
                    index = j;
                }
            }
        }

        visited[index] = 1;
        total_movement += abs(head - requests[index]);
        head = requests[index];

        printf("%d ", head);
    }

    printf("\nTotal Head Movement = %d\n", total_movement);

    return 0;
}
```

Output :

```
Student@DESKTOP-JJ8CFNB MINGW64 ~
$ ./sstf
Order of servicing:
41 65 67 98 122 124 183 14
Total Head Movement = 323
```